Design Details Document

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Design of Defensive Robot (DR):

Specifications:

Length	**978mm (Full extension)**
Width	**953mm (Full extension)**
Height	**711mm (Full extension)**
Weight	**20.5 Kg (Approximately)**

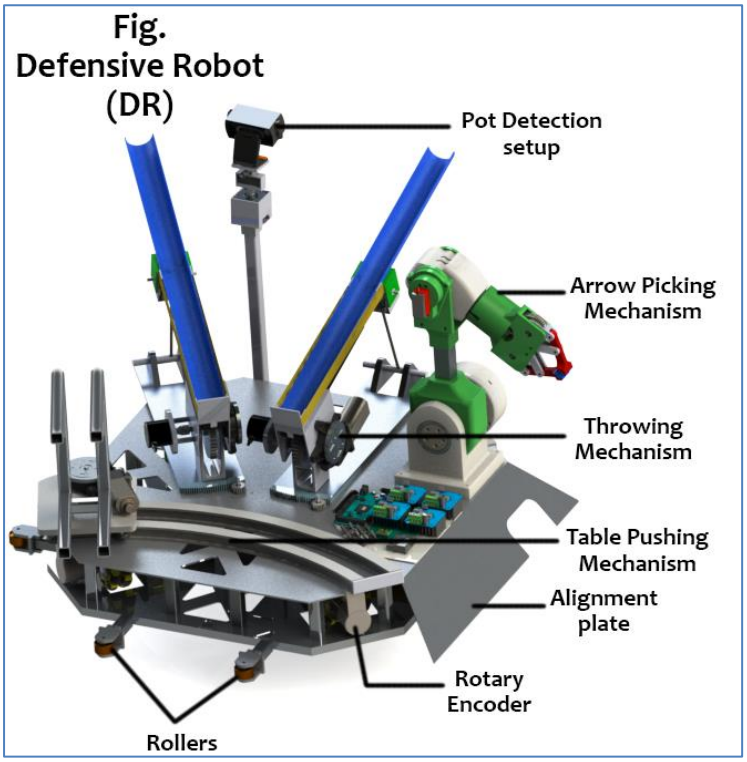


Fig. Defensive Robot (DR)

Type of Drive (4-wheel holonomic Omni drive):

The **holonomic cross drive with Omni wheels** is selected for its capability to move to any coordinate in the X-Y plane by calculating the instantaneous resultants of all 4 wheels, while also providing control over the rotation about the Z-axis. Thus, the Holonomic Omni cross drive has an advantage in terms of maneuverability and traction. The Defensive Robot's chassis is made up of a 6061-grade Aluminum angle beam (12.7mm x 12.7mm x 2mm) which is lightweight, structurally rigid, and ensures sturdiness. Additionally, tiny rollers are fixed on each side of the chassis such that the DR can move using the baffle as a reference for its motion.

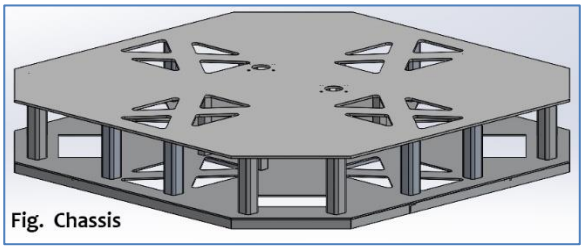
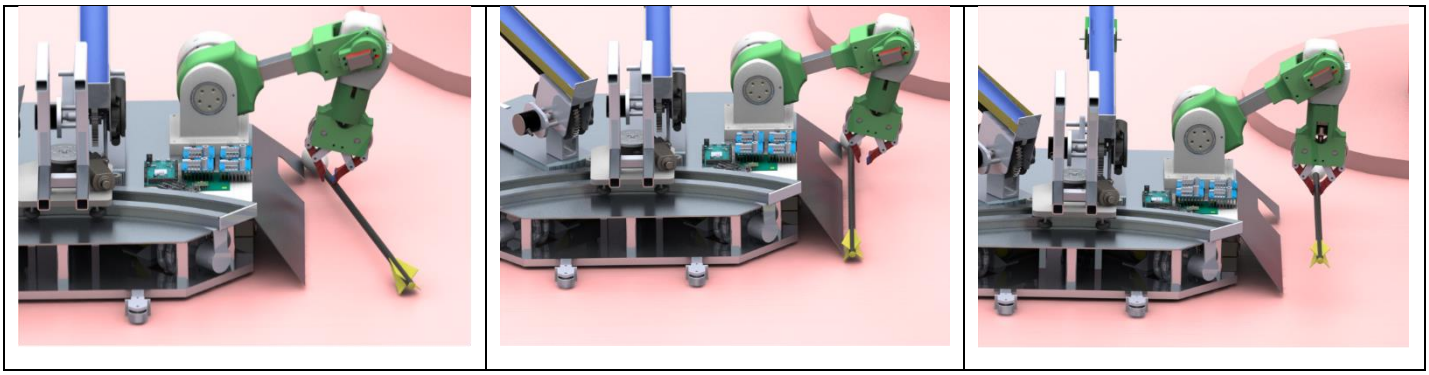


Fig. Chassis

The DR locomotion is semi-autonomous. It can be completely controlled by a manual operator using a PS2 controller and can also be programmed to reach a specified coordinate by pressing a button on the controller. Using encoders on all the wheels and an IMU (Inertial Measurement Unit) at the center of the robot gives an accurate position and orientation of the robot by implementing sensor fusion. Feedback data from these sensors is sent to ROS (Robot Operating System), the software running on a Raspberry Pi, which processes and issues suitable commands to the Arduino Due. Subsequently, Arduino sends appropriate signals to the motor drivers that control the locomotion of the robot.

Arrow picking and passing mechanism:

The arrow picking and passing mechanism consist of 3 Degrees of freedom robotic arm and an alignment plate. Each joint of the arm is controlled by a high-precision servo motor (Stall torque 10 kg-cm). The end effector of this arm has 2 co-planar claws. Since the arrow's body is cylindrical, each claw has a "V" notch along with friction padding for enhanced grip. The end effector has a worm gear meshed with 2 worm pinions that are attached to the claws and are actuated by a servo motor.



To pick an arrow from the ground, the DR uses an alignment plate to push the arrow to align it parallel to the sheet, as shown in **fig**. Since the arrow is now parallel to the end effector gripper, it can easily be grabbed by operating just the base joint and the claws. In this way, the arrow can be picked from the ground irrespective of its orientation. Additionally, if required, the end joint can be rotated to adjust the orientation of the arrow, before passing it to the Throwing Robot (TR).

To pass the arrow, the DR moves toward the grabbing proximity of the TR. After aligning with TR, DR adjusts the joints of the robotic arm and passes the arrow to the grabber clips of the TR.

Design of Throwing Robot (TR)

Specifications:

Length	**954.6mm (Full extension)**
Width	**968mm (Full extension)**
Height	**864.6mm (Full extension)**
Weight	**24.5 Kg (Approximately)**

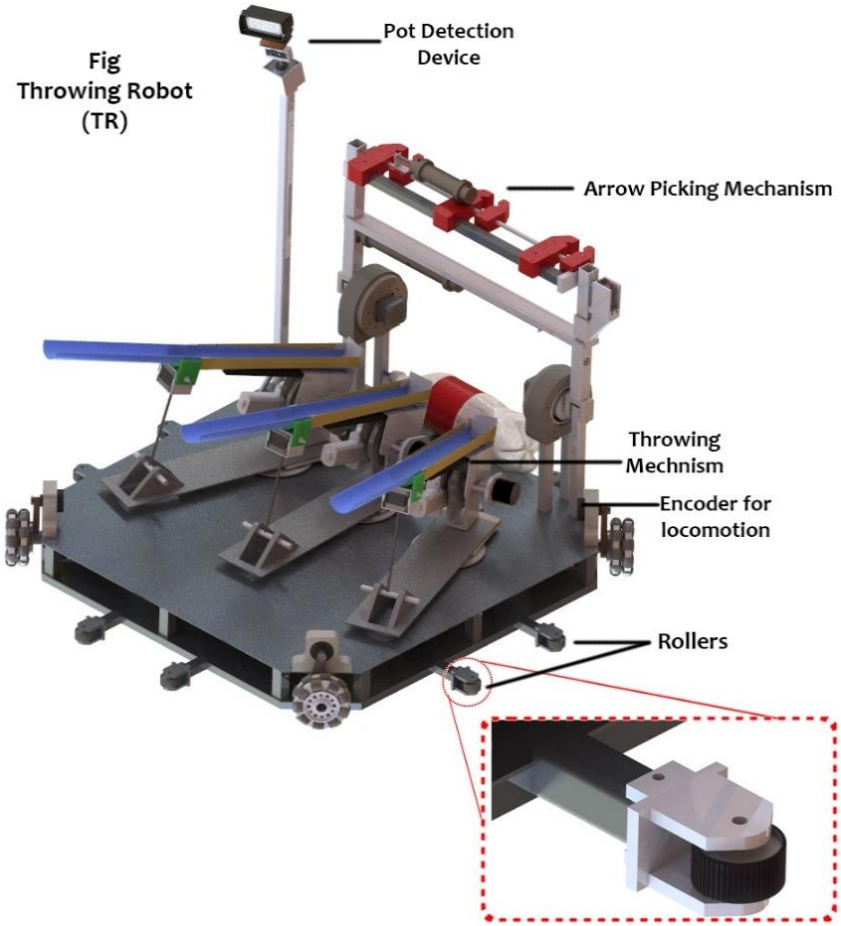


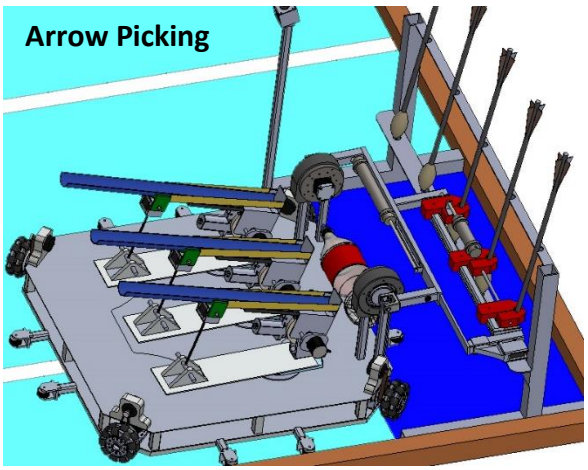
Fig
Throwing Robot (TR)

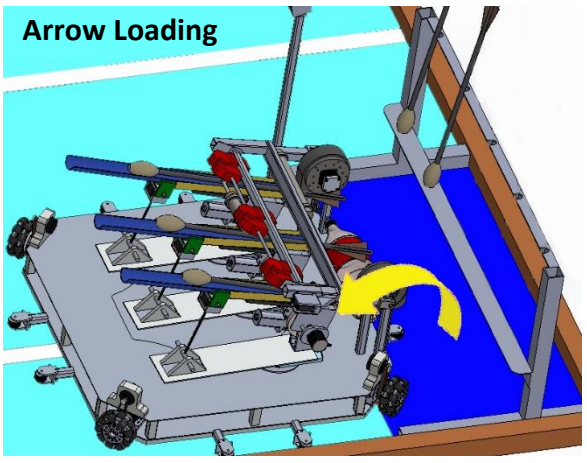
Type of Drive (4-wheel holonomic Omni drive)

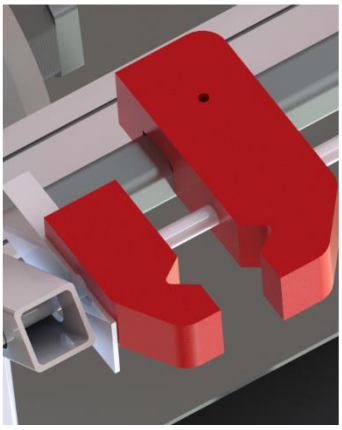
The same holonomic cross drive with holonomic base used in DR is employed for the Throwing Robot (TR). Four Planetary geared motors are used for locomotion. Sensor data from the Arduino DUE is transmitted to Raspberry Pi, the main processor of the system. Odometry is achieved through sensor fusion (i.e., encoders and IMU) using ROS. Here PS2 controller is used for maneuvering. Additionally, rollers are also fixed on the sides of the DR's chassis to assist the wall following with baffles as the reference.

Arrow picking and receiving mechanism

Picking arrows from the rack is done by a mechanism with 2 degrees of freedom. The end effector of this mechanism consists of 3 grabber clips. Each clip has a "V" notch with friction padding for enhanced grip. The clip is made up of two segments, one of which is fixed, while the other is movable. The movable part of each of the grabber clips is connected via a common axis. This common axis is actuated using a pneumatic piston (25mm stroke length), which is responsible for holding an arrow in the "V"notch. These grabbers are mounted on a dynamically balanced frame which is held with a pivot joint attached to a servo motor.







To pick and load arrows from the arrow rack, the angular displacement of the grabber mechanism is needed. This requires high torque which is achieved by using a cycloidal reducer mounted onto each of 2 NEMA-17 stepper motors (Stall torque: 3.2 kg-cm). In order to reach the furthermost arrows on the rack, the grabbers can slide laterally to align themselves with the arrows by using a pneumatic actuator. After grabbing the arrows, they are then aligned back to the launcher to complete the loading.

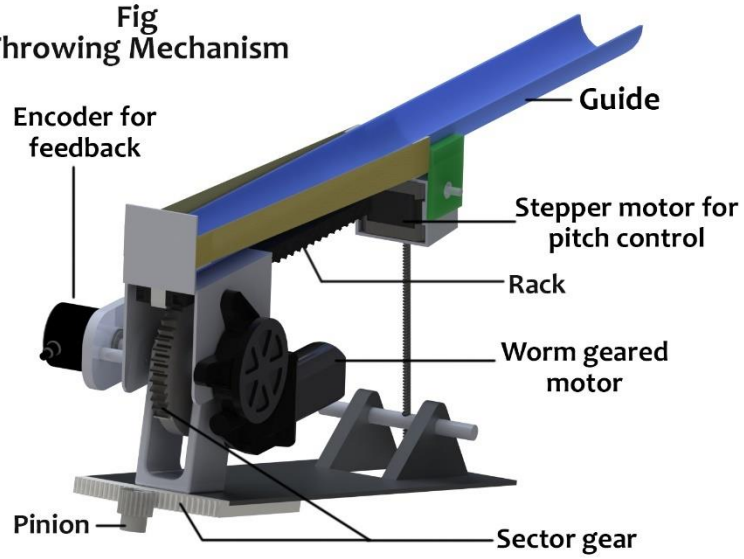
To load the arrows, the servo motor of the balanced frame rotates the pivot joint, thereby rotating arrows and bringing the head upwards and then repositions parallel to the launcher.

The same mechanism is used to receive the arrow from DR, the only difference being that the arrow is received from DR's passing mechanism instead of the arrow rack.

Throwing Mechanism

The throwing mechanism comprises multiple Thera-Band layers, rack, guide, and a sector gear (some numerical value). The Thera-Band is attached to the rack and because of its elastic property (Y = 2.75 MPa), it stores potential energy when it is pulled by a worm gear motor-driven sector gear. This stored energy is then released when the gear reaches the required position on the rack, providing enough energy for the thera-band to launch the arrow through the guide.

Fig
Throwing Mechanism



Projectile equations are used to determine the control angles of the throwing mechanism i.e., the yaw and pitch while the launching speed is determined using the elasticity equations of the thera-band. This ensures that the arrows land into the pot with precision. The destination pot is detected using the camera by color detection algorithm while the accurate distance to the detected pot from the launcher is determined using a LiDAR sensor.

Considering Y= 2.75 *Mpa*
A= (6) x (0.63×10^{-3}) x (30×10^{-3}) m^2

∵The system has 6 Thera-Band layers
$\Delta l = 0.237\ m$, $l_i = 0.35\ m$,

$$v = \sqrt{\frac{Y.A.\Delta l^2}{l_i.m}}$$

$$\Rightarrow v = 9.6m/s \simeq 10m/s$$

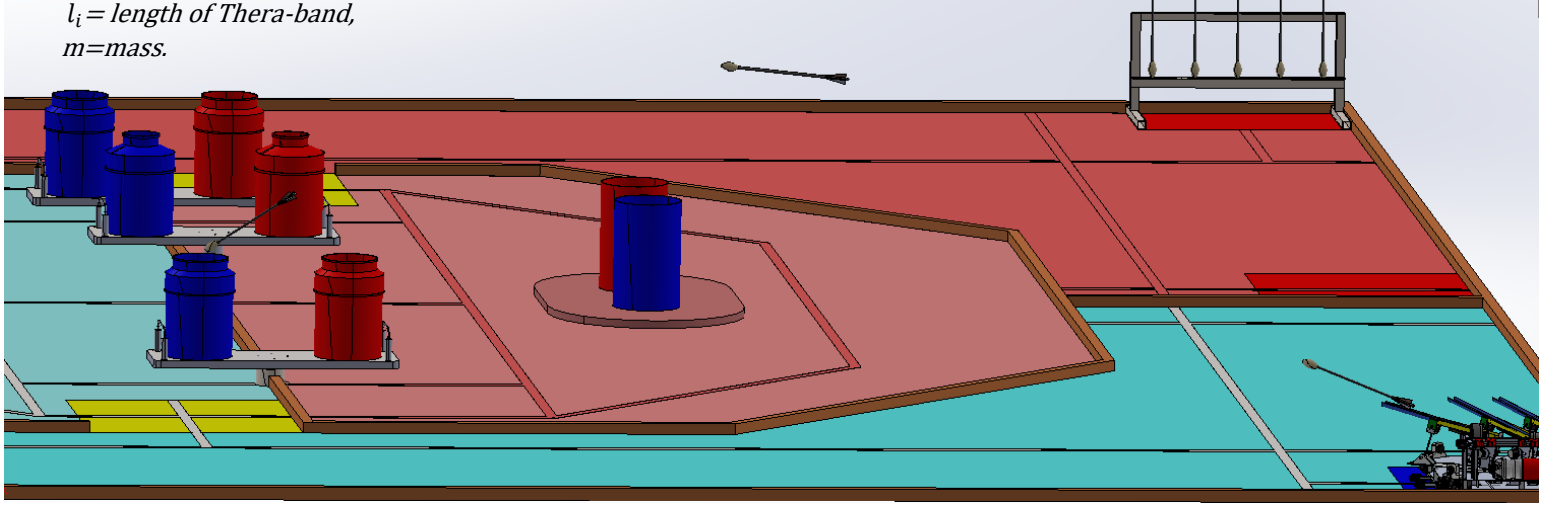
Where Y= Young's modulus,
A=cross sectional area,
Δl = change in length,
l_i = length of Thera-band,
m=mass.

From the Elasticity equations $u = 10m/s$

$$x = u\cos\theta.t \quad \Rightarrow \quad t = \frac{x}{(u\cos\theta)}$$

$$\Rightarrow y - y_0 = u\sin\theta.t - \frac{1}{2}gt^2$$

$$\Rightarrow y - y_0 = u\sin\theta.\frac{x}{(u\cos\theta)} - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2$$



$$\Rightarrow y - y_0 = usin\theta.\frac{x}{(ucos\theta)} - \frac{1}{2}g\left(\frac{x}{ucos\theta}\right)^2$$

$$\Rightarrow y - y_0 = x.tan\theta - \frac{1}{2}g\left(\frac{x}{ucos\theta}\right)^2$$

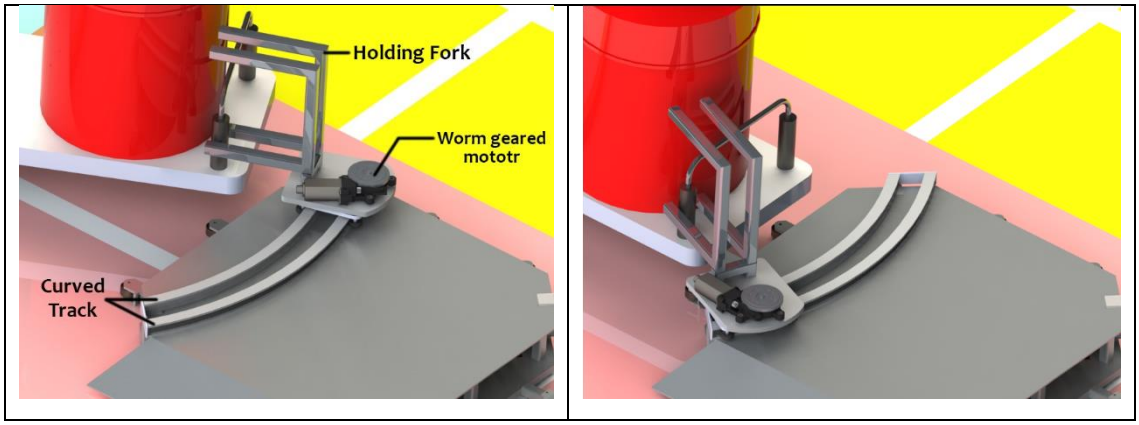
$$\theta = \frac{tan^{-1}\left(\frac{y-y_0}{x}\right) + sin^{-1}\left(\frac{gx^2 + (y-y_0)v^2}{v^2\sqrt{x^2 + (y-y_0)^2}}\right)}{2}$$

where θ = Launching Angle,
y=height of the destination pot, y_0= launching height,
g= acceleration due to gravity, x=Horizontal Range
t= time of flight= launching velocity

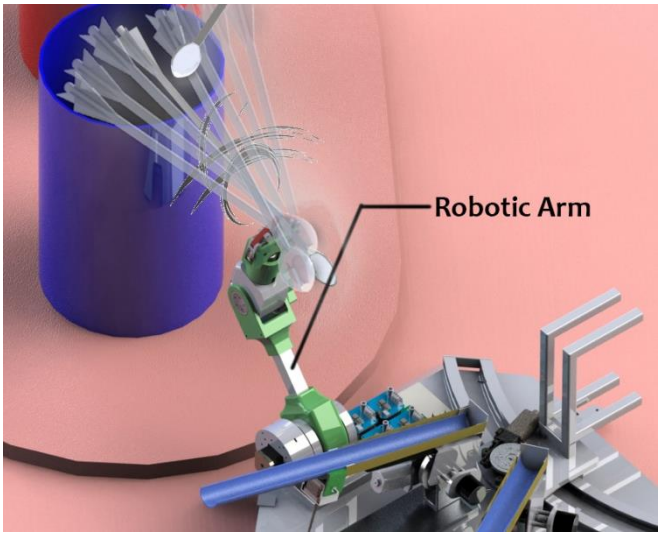
Thus, the horizontal range (x) to be achieved by the arrow's projectile is obtained. To ensure that the landing angles (θ) are maintained around 45°$\pm$10° for all pots, the launching velocity of the arrow is calculated using the equations considering Young's modulus of the Thera-band 2.75 Mpa. Once the velocity is determined, launching angles i.e., the pitch is evaluated using the projectile equations as shown below

In order to achieve the faster twinning, 3 launchers on TR and 2 launchers on DR are fabricated, such that each mechanism can throw the arrow in any of the pots, considering the point “Two robots can throw arrows into the same pot at the same time” mentioned in Q2.4-19 of the official FAQ dated 15/01/2021.

Table Pushing Mechanism and Arrow Interception Mechanism



Pushing/turning:
The table pushing mechanism has 2 forks that are driven by a worm gear motor over a curved track as shown in **fig**. Each holding fork is a 2-pronged structure that can hold a table’s handle. They are designed in such a way that each of the forks can hold/move either type-II or type-III tables, placed at different heights. The mechanism moves on the curved track with the help of a stepper pulley that is attached to the shaft of the worm gear motor and a belt that is fixed to the ends of the outer rail. The structure also has rollers on its base to support its movement over the track. The setup holds the table’s handle in such a way that when the pulley rotates, the entire mechanism slides over the curved track and moves the table accordingly.



Interception: The arrow interception mechanism uses the same robotic arm that was used to pick and pass arrows to the TR. The arm is fixed on the DR chassis in such a way that it can pick an arrow from the ground and also intercept the opponent's incoming arrows. To achieve this, the arm holds an arrow in the “V” notch of the claw, which has friction padding that ensures that the arrow doesn't slip out from its grip. By rotating the end joint of the arm, the arrow held by the claw can be waved, thereby intercepting the opponent's arrow. The maximum height up to which the end effector with an arrow in its claw can reach is 800 mm, which is enough to deflect any arrow coming in the proximity of the pot.

The Robotic arm is chosen so that the DR can perform multiple tasks like picking, passing, and intercepting arrows, using one mechanism making it less complex and reducing the hardware on the robot. The arm is so programmed that it is semi-autonomous and can reach a predefined position with an orientation vector to perform these tasks.

List of Actuators and sensors integrated in TR and DR:

Actuators used:

S.No.	Component	Specifications	Required Power	TR/DR	Usage
1.	Planetary geared motor	350 RPM, 24V DC, 8 kg-cm (stall torque), 84W (Rated power)	38W	**TR & DR**	**Locomotion:** 4 motors attached to the wheels of each robot
2.	Worm Geared (Power Window) motor	100RPM 12VDC, 2.9N-m (Stall Torque) 180W (Rated Power)	48W	**TR & DR**	**Table pushing:** To actuate the pushing mechanism on DR **Throwing:** To control the rotation of sector gear on each launcher
3.	Stepper Motor (Nema-17)	1.2A/Phase (Supply Current), 4.2N-m (Holding Torque), 14.4W /Phase (Rated Power)	8.4 W/Phase	**TR & DR**	**Throwing:**2 steppers to adjust yaw and pitch of each launcher 1 stepper to adjust LiDAR and laser on each robot **Arrow Interception:** To actuate the base joint of the robotic arm **Arrow picking and receiving:** 2 steppers for angular displacement of the grabber
4.	Servo Motor MG995R	5V DC, 10 kg-cm (Stall Torque),	10W	**TR & DR**	**Arrow picking and passing:** 3 servo motors to manipulate the joints of the robotic arm **Arrow picking and receiving:** To rotate the pivot joint
5.	Pneumatic Piston with solenoid	12V DC, 25 mm stroke length.	1.2 W	**TR**	**Picking & Receiving:** one to control the grabber clips and another one to slide the grabber to align it with the furthermost arrow on the rack

Sensors used: All the sensors mentioned in the table are used in both TR and DR

S. No	Component	Specifications	Usage
1.	Optical Rotary Encoder	Incremental type, 5-24V DC, PPR=360, CPR=1440.	**Locomotion**: Attached to all 4 wheels of both the robots to estimate their position & orientation **Throwing:** Providing feedback to the worm geared motor of each launcher
2.	IMU MPU-6050	3-axis gyroscope and accelerometer.	**Locomotion**: To estimate the orientation, acceleration & velocity of each robot
3.	Magnetic Rotary encoder	12 Bit Precision Magnetic Induction, 3.3V DC, CPR=4096.	**Throwing:** To ensure the control angles (i.e., yaw and pitch) of launchers are maintained
4.	TF mini-LIDAR	4.5-6V DC, operating range is 0.3-12 meters.	**Throwing**: To determine the accurate distance from the launcher to the pot. One LiDAR on each robot.
5.	Camera	5VDC, 720p,60fps, 78°FoV	**Throwing:** Attached at the front each robot for pot detection